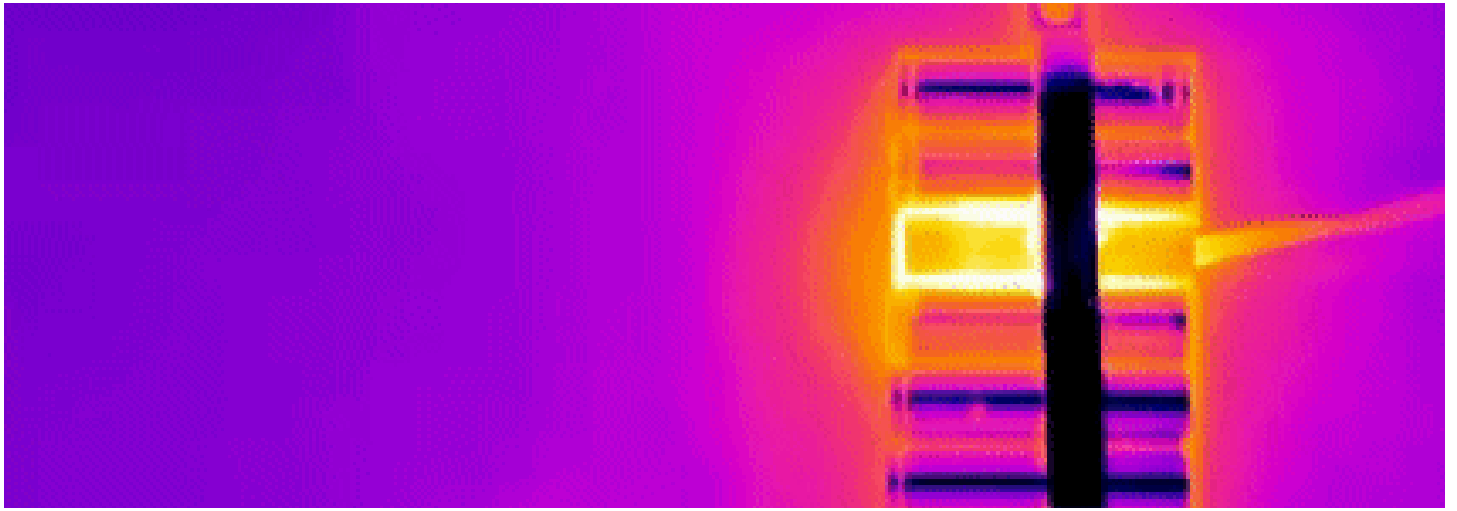




Understanding Thermal Runaway in Lithium-Ion Batteries



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Introduction

Lithium-ion batteries are essential to our everyday lives. They power our smartphones, laptops, electric cars, and numerous other devices. While these batteries are efficient and reliable, they come with a significant risk: thermal runaway. This phenomenon can lead to battery fires or explosions, posing serious safety hazards.

Let's explore what thermal runaway is, why it happens, and how it affects the devices we rely on.

What is Thermal Runaway?

Thermal runaway is a chain reaction within a battery that leads to a rapid increase in temperature. This process can cause the battery to catch fire or even explode. It starts when a part of the battery gets too hot, which causes the chemical reactions inside to accelerate. As the reactions speed up, they produce more heat, creating a vicious cycle that can quickly spiral out of control.